



Warsaw University of Technology

Faculty of Mathematics and Information Science

Data Exploration and Visualisation Project

Outlier Detection

Author: Mateo Jakšić

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1. Introduction

Outlier detection is an important problem in data analysis and machine learning, where the objective is to identify observations that differ significantly from the majority of the data. Such anomalies may correspond to fraud, system failures, measurement errors, or other rare events, making their detection valuable in many practical applications. Since labeled anomalies are often unavailable, unsupervised methods play a key role in this domain.

This project evaluates and compares five unsupervised outlier detection algorithms: One-Class SVM, Isolation Forest, Local Outlier Factor, DBSCAN and Deadwood. The experiments are conducted on multivariate real-world datasets from the ODDS repository and two-dimensional synthetic datasets from the Clustering Benchmark Suite, allowing both quantitative evaluation and visual inspection of the results.

The study focuses on comparing the detection performance of the investigated methods, analyzing their sensitivity to hyperparameter selection and examining how dataset characteristics such as dimensionality and outlier distribution influence their effectiveness. The overall goal is to identify the strengths and weaknesses of each approach and provide practical insights into their application for anomaly detection tasks.

2. Datasets

2.1. Multivariate Datasets

Four datasets were chosen from the Outlier Detection DataSets (ODDS) repository to cover a wide range of dimensionalities, dataset sizes, and contamination rates. Chosen datasets can be seen in Table 1.

Dataset	Points	Dimensions	Outliers	Outlier %
cardio	1831	21	176	9.6%
ionosphere	351	33	126	35.9%
speech	3686	400	61	1.6%
vowels	1456	12	50	3.4%

Table 1: Multivariate datasets

This diversity is intentional. Cardio and Vowels datasets represent typical moderate difficulty datasets. Ionosphere dataset is unusual at 35.9% contamination. Nearly one in three points is an outlier, which challenges methods calibrated for rare anomalies. Speech dataset is the stress test with 400 dimensions and only 61 true outliers in 3,686 points. Any method that performs well here is genuinely robust.

2.2. 2D Datasets

Four 2D datasets were selected from the Clustering Benchmark Suite. Their low dimensionality allows visual evaluation. We can literally see where each algorithm is making mistakes, which is impossible with the multivariate datasets. Chosen datasets can be seen in the Table 2.

Dataset	Points	Outliers	Outlier %
aggregation	788	0	0.0%
flame	240	12	5.0%
ring outlier	1030	30	2.9%

zigzag outlier	280	30	10.7%
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Table 2: 2D datasets

3. Algorithms

3.1. One-Class SVM

One-Class SVM (OC-SVM) is a boundary based anomaly detection method that learns a decision boundary around normal observations. Points located outside this boundary are classified as outliers. The parameter ν controls the expected fraction of anomalies and the complexity of the model. Anomaly scores are based on the distance from the learned boundary.

3.2. Isolation Forest

Isolation Forest detects anomalies by randomly partitioning the data using an ensemble of decision trees. Since outliers are usually easier to separate from the rest of the data, they tend to have shorter path lengths within the trees. The method is computationally efficient and performs well on large datasets.

3.3. Local Outlier Factor

Local Outlier Factor (LOF) identifies anomalies by comparing the local density of a point with the densities of its nearest neighbors. Points that are significantly less dense than their surroundings receive higher anomaly scores. This makes LOF particularly effective for detecting local outliers that may not be globally isolated.

3.4. DBSCAN

DBSCAN is a density-based clustering algorithm that labels points not belonging to any cluster as noise. These noise points are considered outliers. Unlike other methods, DBSCAN produces binary classifications rather than continuous anomaly scores and requires careful tuning of its parameters.

3.5. Deadwood

Deadwood is a graph based anomaly detection method that constructs a minimum spanning tree (MST) from the data. Points connected to the rest of the graph through unusually long edges are identified as outliers. The method is especially intuitive and effective for low dimensional datasets where spatial relationships are easily represented by the graph structure.

4. Results

4.1. Evaluation Metrics

Performance of the algorithms is evaluated based on AUC (Area Under the ROC Curve), Accuracy, Precision, Recall and F1 score.

AUC was used as the primary evaluation metric. AUC measures how well an algorithm ranks anomalous observations above normal ones and is independent of any particular decision threshold. This makes it particularly suitable for comparing methods that generate different types of anomaly scores.

Accuracy measures the proportion of correctly classified observations. While it is easy to interpret, it can be misleading in anomaly detection problems because the majority of data points are usually normal. As a result, a model may achieve high accuracy even if it fails to detect most outliers.

Precision measures the proportion of detected anomalies that are actually true anomalies. High precision indicates a low number of false positives and reflects the reliability of the algorithm's anomaly predictions.

Recall measures the proportion of all true anomalies that are successfully identified by the algorithm. A high recall value indicates that few anomalous observations were missed, making it an important metric when detecting all potential outliers is a priority.

F1 Score was used as a secondary metric. It represents the harmonic mean of Precision and Recall, balancing the trade-off between detecting as many anomalies as possible and minimizing false alarms. The F1 score is especially useful for imbalanced datasets, where anomalies make up only a small fraction of the observations.

4.2. Multivariate Datasets Results

Results for baseline experiment with algorithms on the ODDS datasets are shown in Table 3.

Dataset	Algorithm	AUC	Accuracy	Precision	Recall	F1
cardio	OCSVM	0.870	0.902	0.484	0.256	0.335
	IsoForest	0.927	0.915	0.609	0.318	0.418
	LOF	0.547	0.875	0.217	0.114	0.149
	DBSCAN	0.510	0.114	0.098	1.000	0.178
	Deadwood	0.659	0.852	0.305	0.420	0.353
ionosphere	OCSVM	0.756	0.707	0.897	0.206	0.335
	IsoForest	0.849	0.692	1.000	0.143	0.250
	LOF	0.895	0.692	1.000	0.143	0.250
	DBSCAN	0.540	0.410	0.378	1.000	0.549
	Deadwood	0.810	0.863	1.000	0.619	0.765
speech	OCSVM	0.461	0.928	0.028	0.098	0.043
	IsoForest	0.490	0.937	0.032	0.098	0.049
	LOF	0.508	0.937	0.032	0.098	0.049
	DBSCAN	0.500	0.017	0.017	1.000	0.033
	Deadwood	0.512	0.881	0.020	0.131	0.035
vowels	OCSVM	0.695	0.931	0.160	0.240	0.192
	IsoForest	0.731	0.931	0.151	0.220	0.179
	LOF	0.943	0.950	0.342	0.500	0.407
	DBSCAN	0.500	0.034	0.034	1.000	0.066
	Deadwood	0.923	0.889	0.231	0.960	0.372

Table 3: Multivariate datasets results

The results reveal several important patterns in the behavior of the evaluated anomaly detection algorithms. First, DBSCAN consistently achieved an AUC value of approximately 0.5 across all datasets. This is not necessarily a failure of the algorithm itself but rather a consequence of how AUC is computed. Since DBSCAN produces binary labels instead of continuous anomaly scores, its ROC curve effectively collapses to the diagonal, making AUC an unsuitable evaluation metric. Consequently, F1 score provides a more meaningful assessment of DBSCAN's performance, although the results remained relatively weak with the default parameter settings.

The Speech dataset proved particularly challenging for all methods. None of the algorithms achieved an AUC above 0.512, indicating performance only slightly better than random guessing. This can largely be attributed to the dataset's extremely high dimensionality (400 features), where the distinction between nearby and distant points becomes blurred. As a result, methods that rely on distance or density information struggle to effectively separate normal observations from anomalies.

In contrast, LOF performed exceptionally well on the Vowels dataset, achieving an AUC of 0.943. The dataset's moderate dimensionality and well-defined cluster structure are highly compatible with LOF's local density-based approach. By comparing each point to its nearest neighbors, the algorithm was able to identify anomalies with remarkable accuracy.

Isolation Forest achieved its best results on the Cardio dataset, reaching an AUC of 0.927. This suggests that the anomalous observations in this dataset are globally isolated from the normal data, making them easy to separate through random partitioning. LOF, however, performed poorly on the same dataset, obtaining an AUC of only 0.547. This highlights a limitation of local density methods, although the anomalies are globally unusual, they appear locally similar to nearby observations and therefore are not effectively identified by LOF.

The Ionosphere dataset demonstrated the sensitivity of anomaly detection algorithms to the contamination parameter. Both LOF and Isolation Forest achieved perfect precision, meaning that every detected anomaly was correctly classified. However, their recall was only 0.14, indicating that most anomalies were missed. This behavior can be explained by the mismatch between the default contamination setting of 5% and the dataset's actual contamination rate of approximately 36%, causing the algorithms to label far fewer observations as anomalous than necessary.

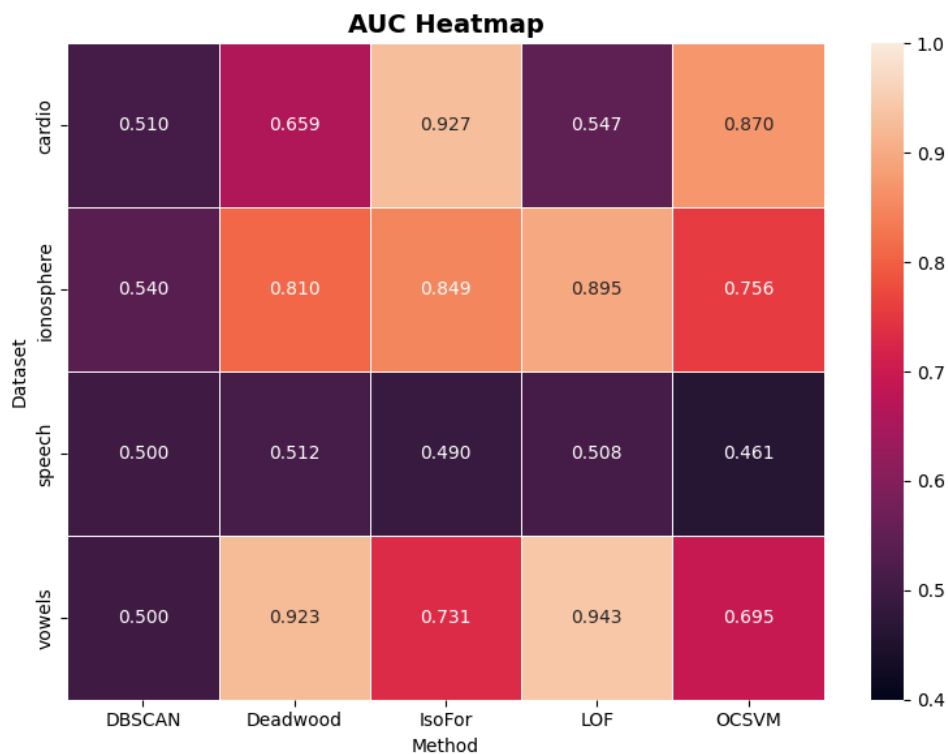


Image 1: AUC heatmap

Image 1 shows the AUC heatmap. It clearly highlights the large variation in algorithm performance across the multidimensional datasets. No single method achieved the best result on every dataset, indicating that anomaly detection performance is strongly dependent on the underlying data structure. Isolation Forest and LOF generally produced the strongest results, but each excelled on different datasets. LOF achieved the highest AUC on the Vowels dataset, while Isolation Forest performed best on Cardio. In contrast, all methods struggled on the Speech dataset, where AUC values remained close to 0.5,

suggesting that the high dimensionality of the data significantly reduced the effectiveness of distance and density based approaches.

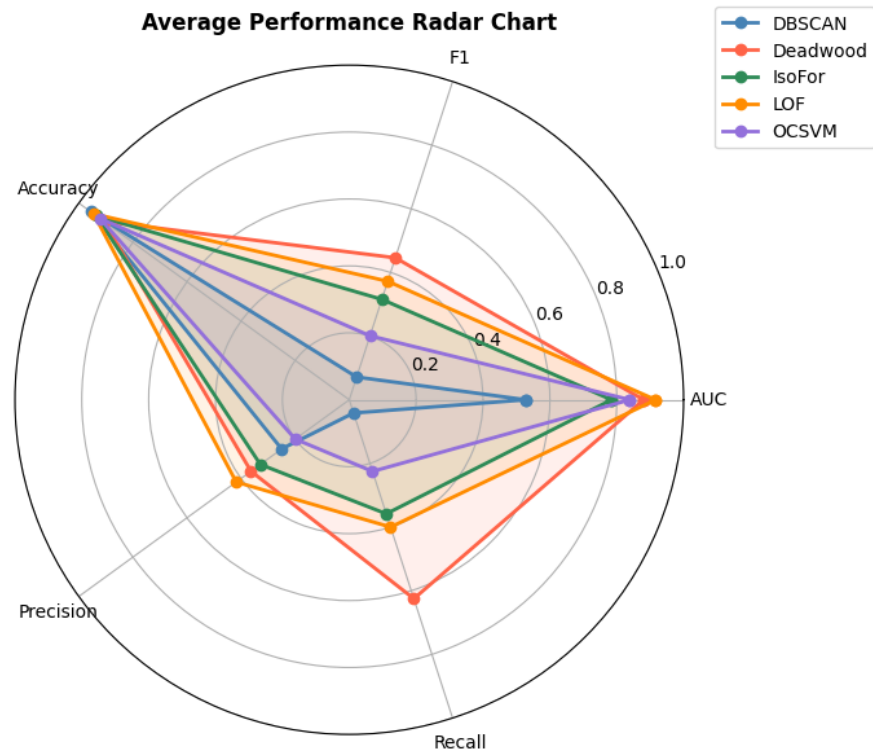


Image 2: Radar graph

The radar graph on Image 2 provides a complementary view by emphasizing the consistency of each algorithm across datasets. Isolation Forest exhibits the most balanced profile, maintaining relatively strong performance on most datasets and avoiding severe failures except on Speech. LOF shows a more irregular shape, reflecting its excellent performance on Vowels but substantially weaker results on Cardio. One-Class SVM achieves moderate results across most datasets without clearly dominating any of them. DBSCAN occupies the smallest area in the radar chart, confirming its poor performance under the default parameter settings and its limited suitability for these datasets without extensive tuning.

4.3. 2D Datasets Results

Results for baseline experiment with algorithms on 2D datasets are shown in Table 3.

Dataset	Algorithm	AUC	Accuracy	Precision	Recall	F1
aggregation	OCSVM	-	0.947	0.000	0.000	0.000
	IsoForest	-	0.949	0.000	0.000	0.000
	LOF	-	0.949	0.000	0.000	0.000
	DBSCAN	-	1.000	0.000	0.000	0.000
	Deadwood	-	0.857	0.000	0.000	0.000
flame	OCSVM	0.745	0.921	0.181	0.167	0.174
	IsoForest	0.385	0.917	0.167	0.167	0.167
	LOF	0.754	0.917	0.167	0.167	0.167
	DBSCAN	0.671	0.958	0.958	0.167	0.286
	Deadwood	0.583	0.825	0.825	0.500	0.222
ring	OCSVM	0.909	0.950	0.302	0.533	0.386
	IsoForest	1.000	0.979	0.577	1.000	0.732
	LOF	0.999	0.979	0.577	1.000	0.732
	DBSCAN	0.500	0.971	0.000	0.000	0.000
	Deadwood	0.985	0.971	0.500	1.000	0.667
zigzag	OCSVM	0.864	0.868	0.316	0.200	0.245
	IsoForest	0.969	0.900	0.929	0.267	0.363
	LOF	0.996	0.936	0.571	0.433	0.591
	DBSCAN	0.500	0.893	0.000	0.000	0.000
	Deadwood	0.986	0.975	0.811	1.000	0.896

Table 1: 2D datasets results

The 2D datasets highlight how strongly anomaly detection performance depends on the geometric structure of the data. In many cases, algorithms succeeded or failed based on

how well their underlying assumptions matched the shape and distribution of the observations.

The Ring dataset produced the strongest result for Isolation Forest, which achieved a perfect AUC of 1.000. The 30 outliers are clearly separated from the circular cluster and can therefore be isolated very quickly by the random partitioning process used by the algorithm. In contrast, DBSCAN performed poorly, obtaining an AUC of 0.5 and an F1 score of 0. With the default parameter settings, the entire dataset was absorbed into a single cluster, resulting in no points being labeled as noise and therefore no anomalies being detected.

On the Zigzag dataset, Deadwood achieved the best overall performance with an AUC of 0.986 and an F1 score of 0.896. The irregular shape of the data presents a challenge for methods that rely on simple boundaries or density estimates. However, Deadwood's graph-based approach is able to follow the natural topology of the zigzag structure through its minimum spanning tree representation, allowing it to identify isolated noise points more effectively.

The Flame dataset proved to be the most difficult of the three datasets. Performance varied considerably across methods, with AUC values ranging from approximately 0.38 to 0.75. The dataset consists of two dense clusters connected by a bridge like structure, while the outliers are located close to the cluster boundaries. As a result, many algorithms interpret these observations as borderline points rather than true anomalies. This suggests that additional domain knowledge or more specialized parameter tuning may be required to accurately detect outliers in datasets with such complex structures.

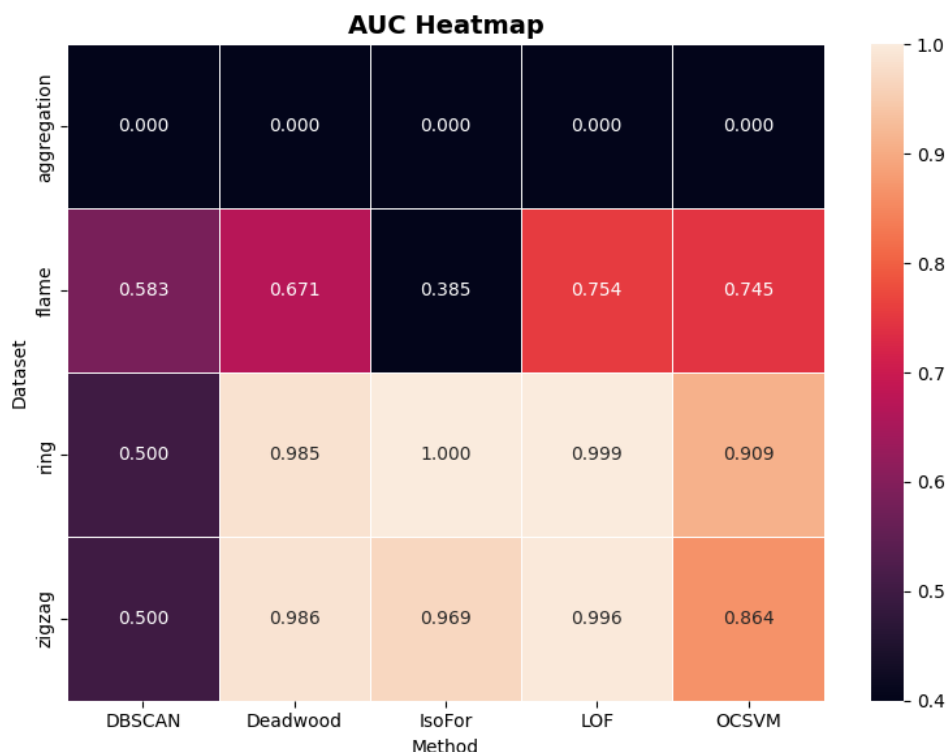


Image 3: AUC heatmap

The AUC heatmap for the 2D datasets in Image 3 shows that performance is largely determined by the geometric structure of the data. Isolation Forest achieved the best result on the Ring dataset (AUC = 1.000), where the outliers are clearly separated from the main cluster and can be easily isolated. Deadwood performed best on the Zigzag dataset, as its graph-based approach effectively captures the irregular shape of the data. In contrast, all methods struggled on the Flame dataset, suggesting that outliers located near cluster boundaries are substantially more difficult to detect than globally isolated anomalies.

The radar chart in Image 4 further highlights the differences between the methods. Deadwood and Isolation Forest cover the largest area, indicating strong overall performance across the 2D datasets. However, their strengths are complementary rather than identical. Isolation Forest excels when anomalies are globally separated from the data, whereas Deadwood performs particularly well on datasets with complex geometric structures. LOF and One-Class SVM show more variable performance, succeeding on

some datasets while struggling on others. DBSCAN exhibits the weakest overall profile, largely due to its sensitivity to parameter settings and its tendency to absorb all points into a single cluster when the default parameters are not appropriate.

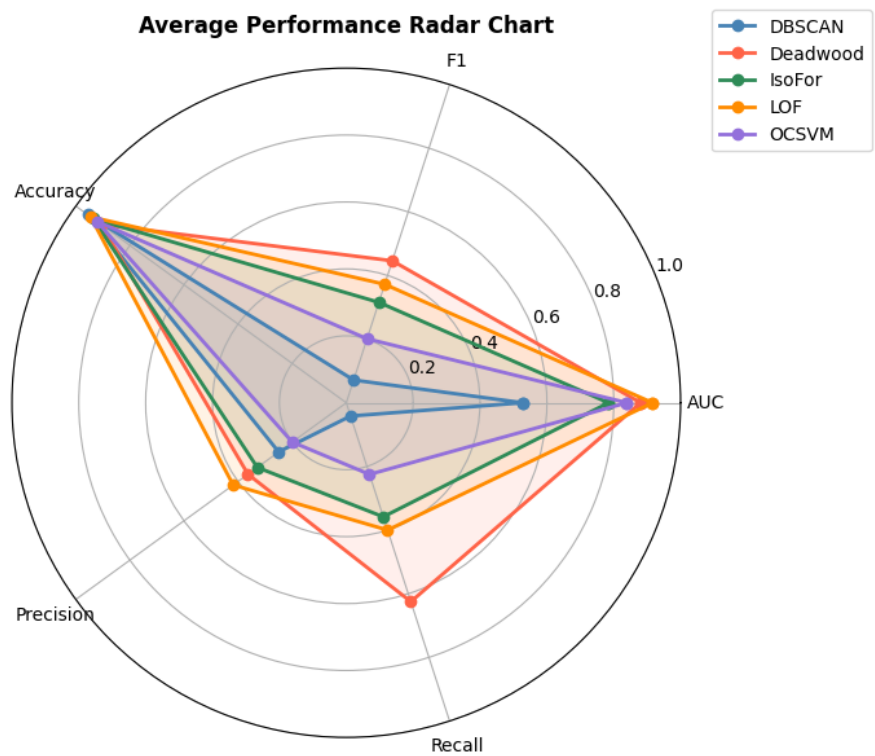


Image 4: Radar graph

Image 5 shows visualization for One-Class SVM algorithm. The algorithm generally identifies observations located outside the main data distribution. It performs well on datasets with clear boundaries but struggles when clusters have complex or irregular shapes.

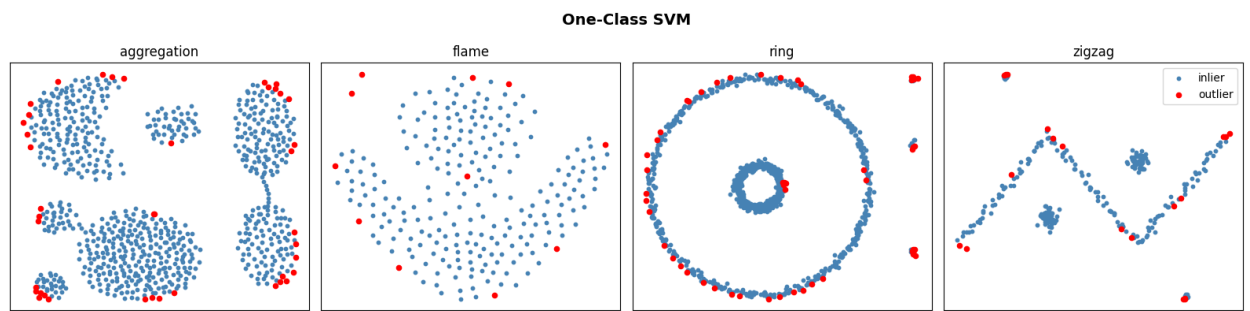


Image 5: Visualization for One-Class SVM

Image 6 shows visualization for Isolation Forest algorithm. The algorithm successfully detects globally isolated observations by separating them through random partitioning. Its results are particularly effective on datasets where outliers are clearly separated from the main clusters.

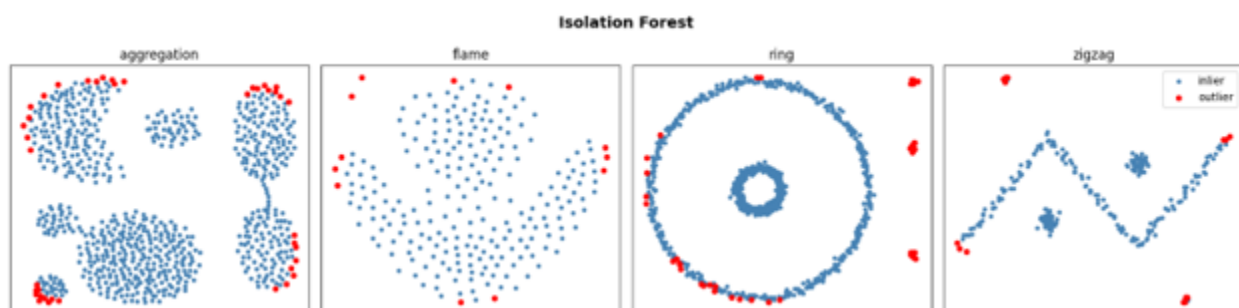


Image 6: Visualization for Local Outlier Factor algorithm

Image 7 shows visualization for Local Outlier Factor algorithm. The algorithm focuses on local density differences and is therefore effective at detecting anomalies relative to their immediate neighborhood. However, it may incorrectly classify boundary points as outliers in datasets with varying densities.

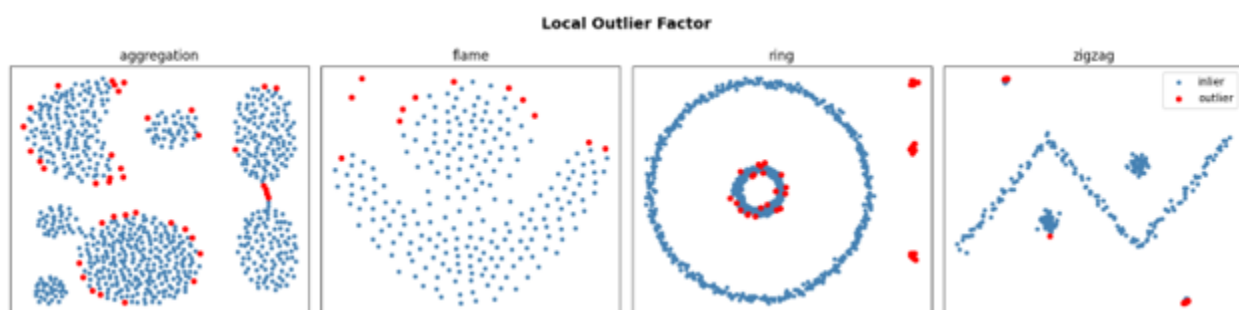


Image 7: Visualization for Isolation Forest algorithm

Image 8 shows visualization for DBSCAN algorithm. The algorithm identifies outliers as noise points that do not belong to any cluster. Its performance is highly dependent on parameter selection, and inappropriate settings may result in either too many or too few detected anomalies.

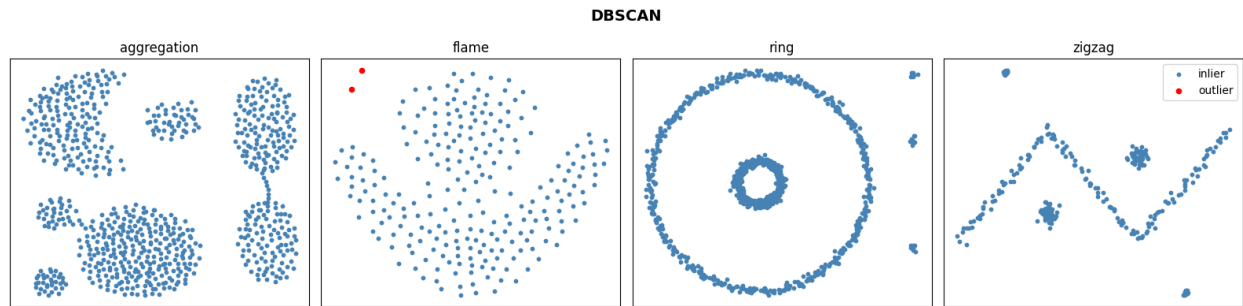


Image 8: Visualization for DBSCAN algorithm

Image 9 shows visualization for Deadwood algorithm. The algorithm uses the structure of a minimum spanning tree to identify observations connected by unusually long edges. This approach is particularly effective for datasets with irregular geometric shapes and complex cluster structures.

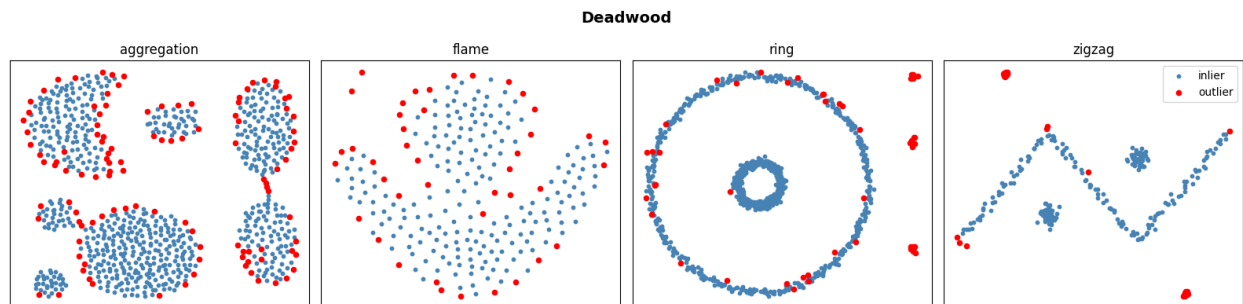


Image 9: Visualization for Deadwood algorithm

5. Hyperparameter Analysis

5.1. One-Class SVM

The hyperparameter analysis shows that the performance of One-Class SVM is strongly affected by the choice of its parameters. Increasing the value of ν generally improves the AUC score for the Cardio, Ionosphere, and Vowels datasets, while the Speech dataset remains largely unaffected, suggesting that the method struggles with this high-dimensional data regardless of the selected value. Among the tested kernels, the RBF kernel consistently achieves the best overall performance, whereas the linear, polynomial, and sigmoid kernels often result in noticeably lower AUC and F1 scores. The γ parameter also plays an important role, with smaller values producing the strongest results and larger values significantly reducing the algorithm's ability to distinguish outliers from normal observations. Overall, the experiments indicate that One-Class SVM is one of the more sensitive methods considered and requires careful hyperparameter tuning to achieve optimal performance.

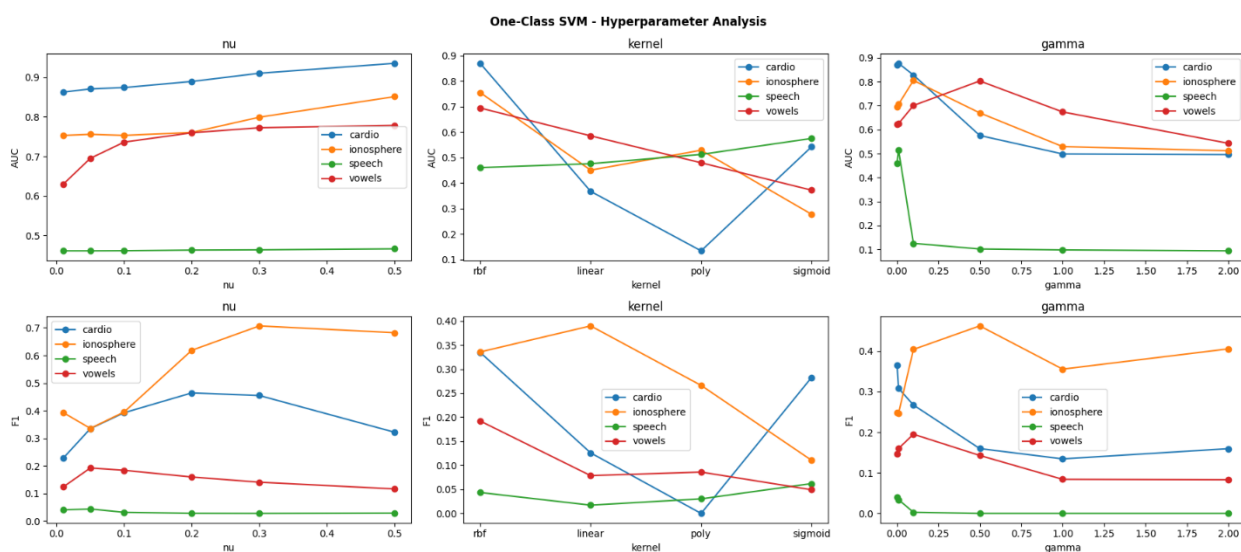


Image 10: Hyperparameter analysis for One-Class SVM

5.2. Isolation Forest

The hyperparameter analysis shows that Isolation Forest is relatively robust to parameter selection. Changing the contamination value has almost no effect on the AUC score across all datasets, although the F1 score improves noticeably for the Cardio and Ionosphere datasets as the contamination parameter approaches the true proportion of outliers. Increasing the number of estimators slightly improves performance on the Cardio and Vowels datasets, while the Speech and Ionosphere datasets remain largely unaffected. Overall, the results suggest that Isolation Forest is one of the most stable methods considered, requiring only limited hyperparameter tuning to achieve good performance.

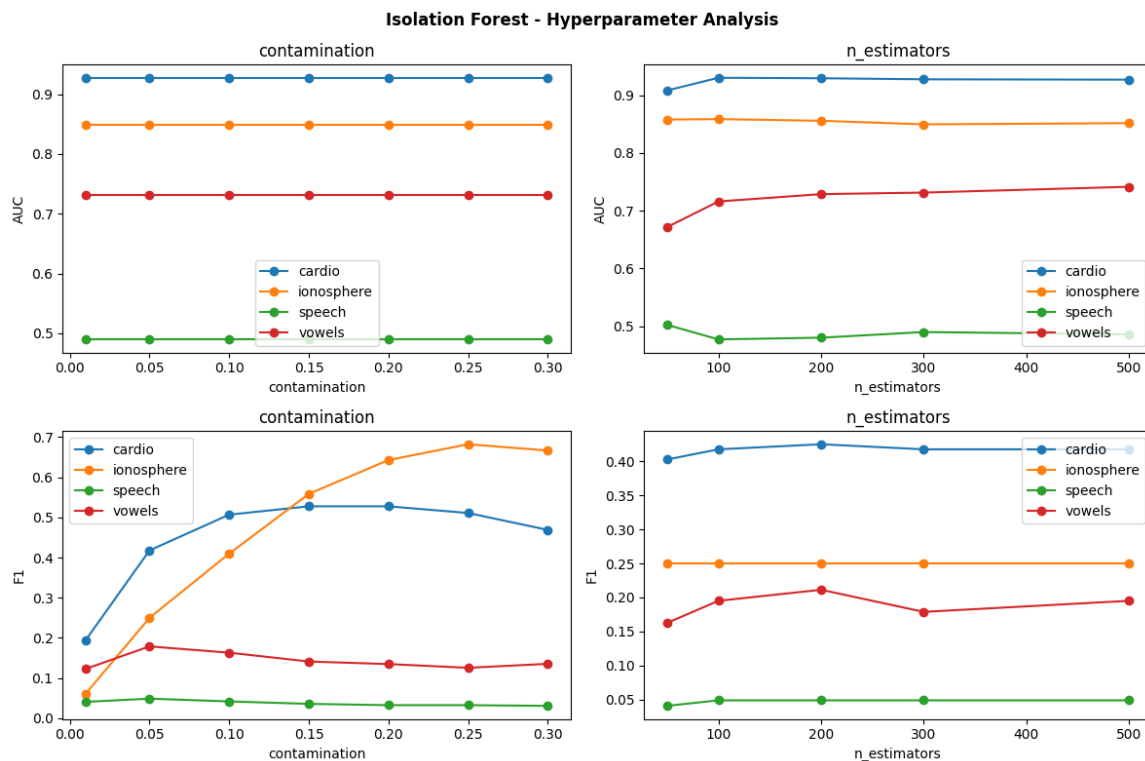


Image 11: Hyperparameter analysis for Isolation Forest

5.3. Local Outlier Factor

The analysis indicates that the choice of number of neighbors has a significant impact on the performance of Local Outlier Factor. Increasing the neighborhood size substantially improves the AUC score for the Cardio dataset, while the Speech dataset exhibits the opposite trend, with performance gradually decreasing as more neighbors are considered. The Ionosphere and Vowels datasets remain relatively stable across the tested range, consistently achieving high AUC values. The F1 score follows a similar pattern, suggesting that LOF is moderately sensitive to the neighborhood parameter and that its optimal value depends strongly on the dataset characteristics.

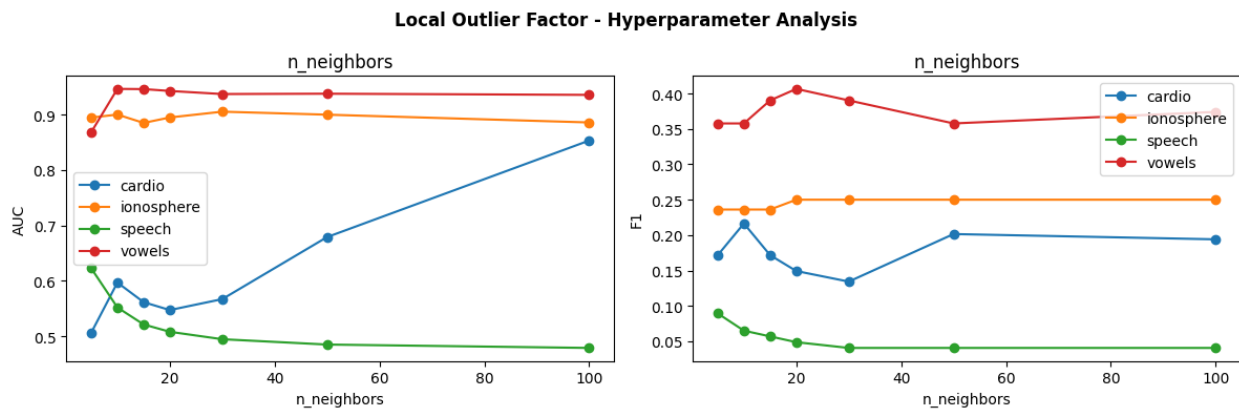


Image 12: Hyperparameter analysis for Local Outlier Factor

5.4. DBSCAN

The hyperparameter analysis reveals that DBSCAN is highly sensitive to the choice of eps. Increasing this parameter leads to considerable improvements in both AUC and F1 score for the Ionosphere and Vowels datasets, with the Vowels dataset reaching its best performance around $\text{eps} = 1.5$. In contrast, the Speech dataset remains almost completely unaffected by changes in eps, indicating that the algorithm struggles with this high-dimensional data regardless of the selected value. The min_samples parameter has only a minor influence on the results, with both AUC and F1 score remaining nearly constant across the tested range. Overall, the experiments suggest that careful tuning of eps is essential for obtaining good performance with DBSCAN.

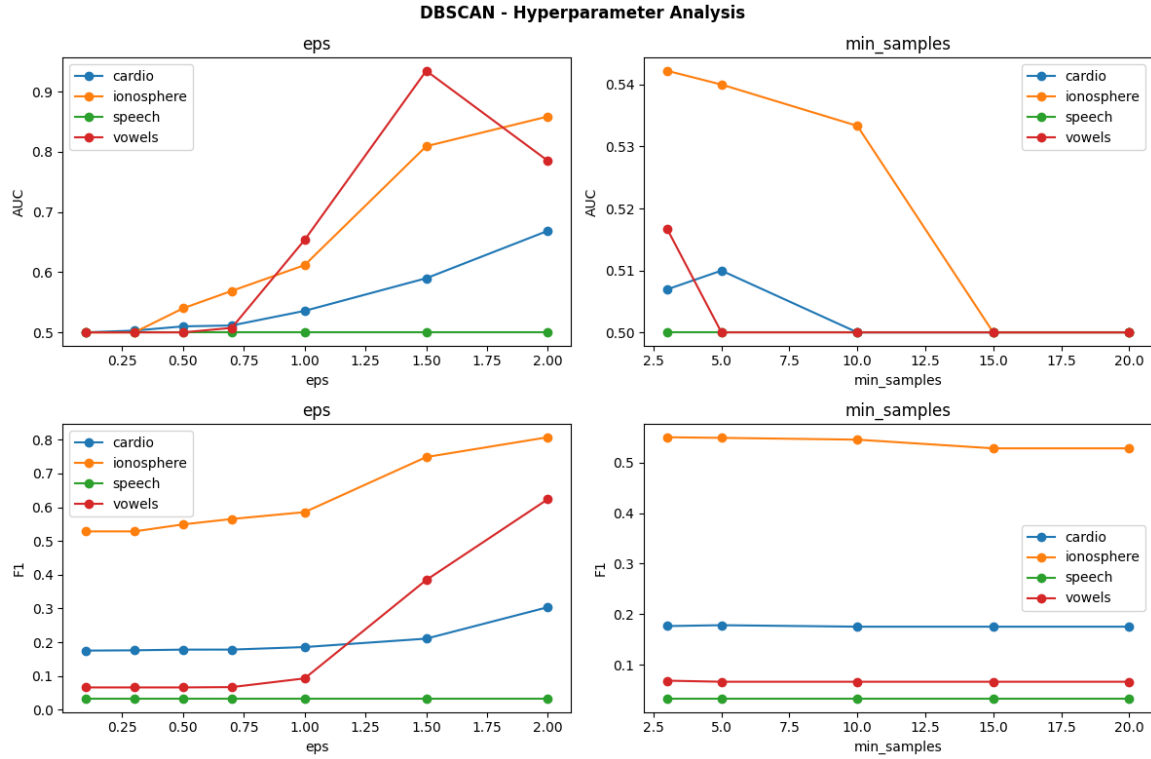


Image 13: Hyperparameter analysis for DBSCAN

5.5. Deadwood

The results show that Deadwood is relatively insensitive to changes in its main hyperparameter M . The Vowels dataset consistently achieves the highest AUC values across all tested settings, while the Speech dataset remains largely unchanged and obtains the lowest scores. For the Cardio dataset, increasing M slightly decreases performance, whereas the Ionosphere dataset experiences only a small gradual decline. Similar trends can be observed for the F1 score, with only minor fluctuations across the tested values. Overall, the experiments indicate that Deadwood is a stable method that provides consistent results without requiring extensive hyperparameter tuning.

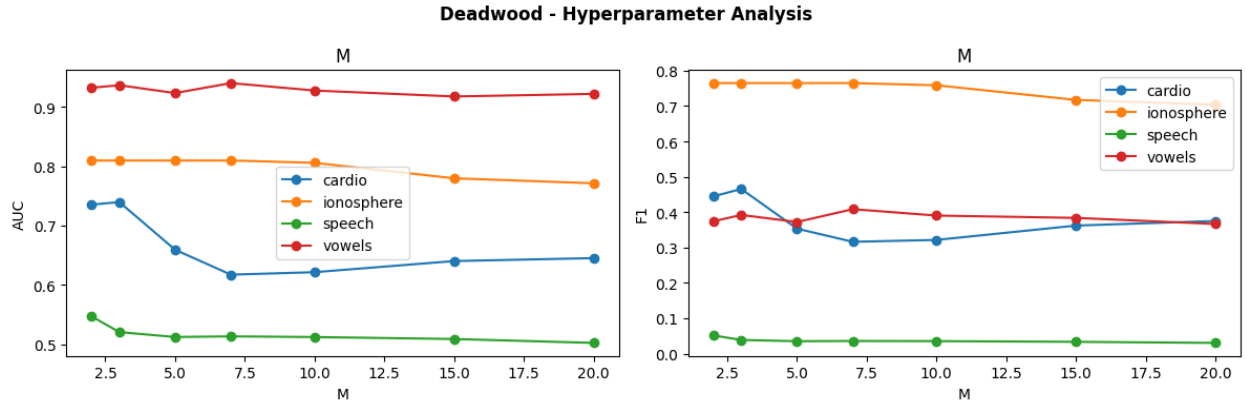


Image 14: Hyperparameter analysis for Deadwood

5.6. Stability analysis

The hyperparameter analysis reveals noticeable differences in the stability of the investigated methods. Isolation Forest and Deadwood are the most robust algorithms, maintaining consistent performance across a wide range of parameter values and requiring relatively little tuning. Local Outlier Factor exhibits moderate sensitivity, with the number of neighbors having a visible impact on the results, particularly for datasets with different local density structures. In contrast, One-Class SVM and DBSCAN are considerably more sensitive to hyperparameter selection. For One-Class SVM, the choice of kernel and the values of ν and γ can lead to substantial changes in detection quality, while DBSCAN strongly depends on selecting an appropriate ϵ value. Overall, the experiments suggest that Isolation Forest and Deadwood are the easiest methods to apply in practice, whereas One-Class SVM and DBSCAN require more careful tuning to achieve optimal performance.

6. Robustness and Failure Case Analysis

6.1. Effect of Dimensionality

The experiments show that dataset dimensionality has a considerable impact on algorithm performance. The Speech dataset, containing 400 features, proved to be the most challenging. Algorithms achieved relatively poor results on this dataset regardless of hyperparameter selection, suggesting that these methods struggle in very high-dimensional spaces. In contrast, Isolation Forest and Deadwood maintained relatively stable performance, indicating greater robustness to increasing dimensionality.

6.2. Effect of Dataset size

The investigated datasets vary substantially in the number of observations, ranging from 351 samples in Ionosphere to 3686 samples in Speech. Isolation Forest benefited from larger datasets, as its ensemble-based approach was able to build more representative random partitions. One-Class SVM and LOF also performed well on medium-sized datasets such as Cardio and Vowels, but their performance became less consistent on datasets with either very few or very many observations.

6.3. Effect of Outlier Distribution

The methods differ in the types of anomalies they are best suited to detect. Local Outlier Factor is specifically designed to identify observations with unusually low local density, making it effective for detecting local outliers. Isolation Forest and One-Class SVM are better suited for identifying global anomalies that are clearly separated from the majority of the data. DBSCAN performs well when outliers lie outside dense clusters, but may struggle when clusters have varying densities or when anomalies are embedded within larger structures.

7. Discussion

The experiments demonstrate that the performance of unsupervised outlier detection algorithms depends strongly on the characteristics of the dataset and the choice of hyperparameters. No single method consistently achieved the best results across all datasets, highlighting the importance of selecting an algorithm that matches the underlying data structure.

Among the evaluated methods, Isolation Forest and Deadwood proved to be the most robust. Both algorithms maintained relatively stable performance across different datasets and showed limited sensitivity to hyperparameter changes. Isolation Forest, in particular, handled the high-dimensional Speech dataset better than the other methods, making it a reliable general-purpose approach.

One-Class SVM was capable of achieving very good results, especially on the Cardio, Ionosphere, and Vowels datasets, but only after careful tuning of parameters such as ν , γ , and the kernel type. The experiments showed that the RBF kernel consistently outperformed the alternatives, while the Speech dataset remained challenging regardless of parameter selection.

Local Outlier Factor performed well when anomalies differed significantly from their local neighborhood, but its effectiveness depended heavily on the choice of the number of neighbors. Similarly, DBSCAN achieved good results on the two-dimensional benchmark datasets with clearly separated clusters, but its performance deteriorated when the data became high-dimensional or when suitable values of ϵ could not be identified.

The hyperparameter analysis further emphasized these differences. Isolation Forest and Deadwood required relatively little tuning, whereas One-Class SVM and DBSCAN were considerably more sensitive to parameter selection. Overall, the experiments suggest that algorithm robustness is just as important as peak performance, particularly when little prior knowledge about the data is available.

8. Conclusion

This project evaluated five unsupervised outlier detection algorithms on both real-world and synthetic datasets. The results showed that no single method is universally superior. Isolation Forest and Deadwood provided the most stable performance across different datasets, while One-Class SVM, Local Outlier Factor, and DBSCAN were more dependent on appropriate hyperparameter selection and dataset characteristics. The experiments also highlighted the influence of dimensionality and outlier structure on detection quality. Overall, the study demonstrates that successful anomaly detection requires both suitable algorithm selection and careful consideration of the underlying data.